WORKED SOLUTION

STEP II 1987, Question 11

Mechanics • Friction and Moments

STEP II 1987, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q11

The key idea

The pivot force is unknown and we do not want it, so we take moments about the pivot O, where it has no moment. The rod is three-dimensional, but the plane of the ring gives two natural directions: perpendicular to the plane (where the normal reaction acts) and within it (where friction acts). Splitting the weight the same way, one moment equation gives the normal reaction and another gives the friction. Their ratio is μ.

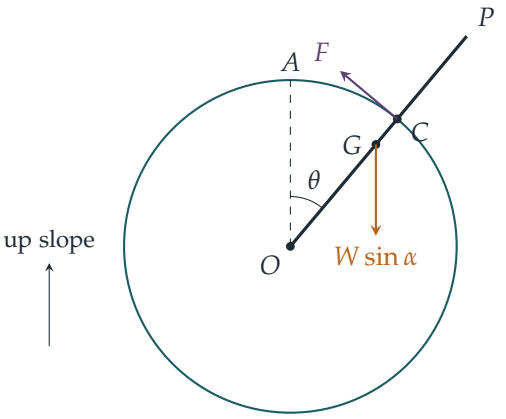
Solution

The model. The rod is a uniform rod OP of weight W, with its centre of mass G at its midpoint, so $OG = \frac{1}{2}b$. It is smoothly pivoted at O, the centre of the ring. It lies across the ring at a point C of the circumference, so $OC = a$. Since O and C both lie in the plane of the ring, the rod lies in that plane. Let A be the highest point of the ring, so OA points straight up the line of greatest slope, and $\angle AOC = \theta$.

The forces. The forces on the rod are:

- its weight W at G;
- the force at the smooth pivot O, which is unknown but has no moment about O;
- the contact force at C. The rod and the ring's wire cross at C. The rod runs along the radius and the wire runs along the tangent, and both lie in the plane, so their common normal is perpendicular to the plane. The normal reaction R is therefore perpendicular to the plane of the ring. Friction F lies in the plane. The rod can only move by turning about O, which sends C along the tangent to the ring. So friction acts along that tangent, perpendicular to the rod.

Resolve the weight into $W\cos\alpha$ perpendicular to the plane (into it) and $W\sin\alpha$ down the line of greatest slope, within the plane. The diagram is drawn looking straight down onto the plane, with A at the top. R points out of the page at C.



Moments about O. Because the pivot is smooth, the total moment about O of the remaining forces is zero about *every* axis through O. Two axes give useful equations.

Axis in the plane, through O, perpendicular to the rod. Only the forces perpendicular to the plane have a moment about this axis. These are $W\cos\alpha$ at distance $\frac{1}{2}b$ and R at distance a. The in-plane forces F and $W\sin\alpha$ meet this axis or are parallel to it, so they have no moment about it. Hence

$$R\,a = W\cos\alpha\cdot\tfrac{1}{2}b, \qquad \text{so} \qquad R = \frac{Wb\cos\alpha}{2a}.$$

Axis perpendicular to the plane, through O. Only in-plane forces matter now. The rod makes angle θ with the up-slope direction, so the component of $W\sin\alpha$ perpendicular to the rod is $W\sin\alpha\sin\theta$, acting at distance $\frac{1}{2}b$. Friction is already perpendicular to the rod, at distance a. Hence

$$F\,a = W\sin\alpha\sin\theta\cdot\tfrac{1}{2}b, \qquad \text{so} \qquad F = \frac{Wb\sin\alpha\sin\theta}{2a}.$$

The in-plane part of the weight tends to swing the rod away from A, down towards the bottom of the ring, and friction opposes this.

? **Ask yourself** Couldn't friction also have a component along the rod?

A force along the rod passes through O, so it has no moment about O. It cannot help or hinder the rotation, and the pivot can supply any force along the rod. The only possible motion is turning about O, so the friction that matters for limiting equilibrium is the component along the tangent, which is F. We take friction to act wholly along the tangent.

Limiting equilibrium. In limiting equilibrium $F = \mu R$:

$$\frac{Wb\sin\alpha\sin\theta}{2a} = \mu\,\frac{Wb\cos\alpha}{2a} \qquad \Longrightarrow \qquad \boxed{\mu = \tan\alpha\,\sin\theta}$$

(we take $0 \le \theta \le \pi$). The rod rests in equilibrium at angle θ whenever $\tan\alpha\sin\theta \le \mu$. Notice that a, b and W have all cancelled.

Checks

- If $\theta = 0$ or $\theta = \pi$, the rod lies along the line of greatest slope. Then the in-plane weight pulls straight along the rod, and no friction is needed: $\mu = 0$. The same holds when $\alpha = 0$, since a horizontal ring gives no in-plane weight.
- Friction is hardest to supply when $\theta = \frac{1}{2}\pi$, with the rod horizontal. Then $\mu = \tan\alpha$, which matches a block on a slope of angle α.
- A numerical check built the forces as 3D vectors for 2000 random choices of α, θ, a, b, W. It solved the vector moment equation about O for R and F, and found F/R agreeing with $\tan\alpha \sin\theta$ to within 4×10^{-15}. For example, $\alpha = 30^\circ$ and $\theta = 60^\circ$ give $\mu = \frac{1}{2}$.

Common slips

- Taking the normal reaction at C to be along the radius, as it would be for a rod resting *inside* a ring in a vertical plane. Here the rod lies across the wire, so the normal is perpendicular to the plane of the ring.
- Resolving forces instead of taking moments. That brings in the three unknown components of the pivot force, which the question doesn't want.

Beyond STEP

Here we took moments about two perpendicular axes through O. This is a shortcut for the single vector equation $\sum \mathbf{r} \times \mathbf{F} = \mathbf{0}$, which says that the total torque about O vanishes. Torque as a vector, and the equilibrium conditions for a rigid body in three dimensions, are standard in first-year mechanics. They lead on to the rotation of rigid bodies about a fixed point. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops torque and angular momentum as vectors in exactly this way.

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